BT614T – INDUSTRIAL BIOTECHNOLOGY

FINAL TERM IMPORTANT QUESTIONS

Prepared by: *Sir Mughal*

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SHORT QUESTIONS

1. In how many ways, a cell can be harvested in fermentation?

Cells can be harvested from fermentation broth mainly by two methods:

- ✓ ***Filtration:*** The cells are separated from the liquid using a filter medium. It is suitable for filamentous fungi and other large microbial cells.
- ✓ ***Centrifugation:*** The fermentation broth is rotated at high speed, causing the cells to settle due to centrifugal force. This method is commonly used for bacteria and yeast because it is rapid and efficient.

2. Write some uses of citric acid in industry.

Citric acid has several industrial uses:

- ✓ ***Food industry:*** Used as an acidulant, flavor enhancer, preservative, and pH regulator in beverages, candies, and processed foods.
- ✓ ***Pharmaceutical industry:*** Used in medicines, syrups, and effervescent tablets due to its acidic and stabilizing properties.
- ✓ ***Cosmetic industry:*** Used in skin-care products as a pH adjuster and antioxidant.
- ✓ ***Detergent industry:*** Used as a chelating agent to remove metal ions and improve cleaning efficiency.
- ✓ ***Industrial applications:*** Used in metal cleaning, textile processing, and as a component in biodegradable products.

3. Enlist the technique to product recovery.

Product recovery techniques used in fermentation include:

- ✓ ***Filtration:*** A method used to separate microbial cells and solid particles from the fermentation broth by passing it through a filter medium.
- ✓ ***Centrifugation:*** Cells and other suspended materials are separated from the broth by applying centrifugal force, causing heavier particles to settle.
- ✓ ***Extraction:*** The desired product is separated from the fermentation broth using suitable solvents based on its chemical properties.

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- ✓ ***Precipitation:*** The product is converted into an insoluble form by adding chemicals or changing conditions, allowing it to be collected easily.
- ✓ ***Chromatography:*** A purification technique used to separate and obtain highly pure products based on differences in their chemical properties.
- ✓ ***Crystallization:*** The purified product is converted into crystals from a solution for easy recovery and storage.

4. What are some healthcare products of microbial fermentation?

Healthcare products obtained through microbial fermentation include:

- ✓ ***Antibiotics:*** Produced by microorganisms and used to treat bacterial infections, e.g., penicillin and streptomycin.
- ✓ ***Vaccines:*** Some vaccines are produced using microbial fermentation techniques.
- ✓ ***Vitamins:*** Microorganisms are used for the production of vitamins such as vitamin B12.
- ✓ ***Hormones and therapeutic proteins:*** Fermentation is used to produce products like insulin and other recombinant proteins.
- ✓ ***Amino acids:*** Compounds such as glutamic acid are produced and used in pharmaceutical preparations.
- ✓ ***Enzymes:*** Microbial enzymes are used in diagnostic kits and medical treatments.

5. Enlist three compounds of photosynthesis in eubacteria.

Three compounds involved in photosynthesis in eubacteria are:

- ✓ ***Bacteriochlorophyll:*** A photosynthetic pigment that absorbs light energy for photosynthesis in photosynthetic bacteria.
- ✓ ***Carotenoids:*** Accessory pigments that help absorb light and protect cells from photo-oxidative damage.
- ✓ ***Phycobilins:*** Water-soluble pigments found in cyanobacteria that capture light energy and transfer it to chlorophyll.

6. Write example of two GRAS bacteria, yeast and fungi.

Examples of GRAS (Generally Recognized As Safe) microorganisms are:

- ✓ ***Bacteria:*** *Lactobacillus acidophilus, Bacillus subtilis*
- ✓ ***Yeast:*** *Saccharomyces cerevisiae*
- ✓ ***Fungi:*** *Aspergillus oryzae, Penicillium chrysogenum*

7. Define autotrophs. Write its types and define them.

Autotrophs:

Autotrophs are organisms that can synthesize their own food from simple inorganic substances such as carbon dioxide and water using energy from sunlight or chemical reactions.

Types of autotrophs:

- ✓ ***Photoautotrophs:*** These organisms use light energy to convert carbon dioxide into organic compounds through photosynthesis. Example: cyanobacteria, green algae.
- ✓ ***Chemoautotrophs:*** These organisms obtain energy by oxidizing inorganic substances such as ammonia, sulfur, iron, or hydrogen and use it to fix carbon dioxide. Example: *Nitrosomonas* and *Thiobacillus*.

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8. How a cell can be extracted from broth?

Cells can be extracted from fermentation broth by the following methods:

- ✓ ***Filtration:*** The broth is passed through a filter that retains microbial cells while allowing the liquid to pass through.
- ✓ ***Centrifugation:*** The broth is rotated at high speed, causing cells to settle at the bottom due to centrifugal force, forming a cell pellet.
- ✓ ***Sedimentation:*** Cells are allowed to settle naturally under gravity and are then collected.

9. What is mutagen? Give two examples.

A mutagen is a physical or chemical agent that causes changes or mutations in the genetic material (DNA) of an organism. Mutagens are used in microbial strain improvement to develop organisms with desirable characteristics for industrial production.

Examples:

1. ***Ultraviolet (UV) radiation*** – a physical mutagen that causes DNA damage.
2. ***Ethyl methanesulfonate (EMS)*** – a chemical mutagen that induces genetic mutations.

10. Why antifoam is used in fermentation process?

Antifoam is used in fermentation processes to prevent and control excessive foam formation in the fermenter. During fermentation, foam is produced due to the presence of proteins, carbohydrates, and other surface-active substances along with continuous aeration and agitation.

Excessive foam can cause several problems such as contamination risk, blockage of air filters, reduction in oxygen transfer, interference with sensors, loss of culture medium, and decrease in fermenter efficiency. Antifoaming agents reduce surface tension and break down foam bubbles, helping to maintain proper fermentation conditions and improve product yield.

11. Which bacteria produces glutamic acid?

Glutamic acid is mainly produced by the bacterium *Corynebacterium glutamicum*. It is an important industrial microorganism used for the large-scale production of L-glutamic acid, which is used in the food industry for manufacturing monosodium glutamate (MSG) and as a flavor enhancer. Other bacteria such as *Brevibacterium flavum* and *Brevibacterium lactofermentum* can also produce glutamic acid.

12. What are the form(s) of RNA for protein synthesis?

Three forms of RNA are involved in protein synthesis:

- ✓ ***Messenger RNA (mRNA):*** Carries genetic information from DNA to ribosomes and acts as a template for protein formation.
- ✓ ***Transfer RNA (tRNA):*** Transports specific amino acids to the ribosome and matches them with the codons on mRNA during protein synthesis.
- ✓ ***Ribosomal RNA (rRNA):*** Forms the structural and catalytic component of ribosomes, where protein synthesis takes place.

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13.How many ways of cell cultivation are there?

Cell cultivation is mainly carried out in three ways:

- ✓ ***Batch culture:*** Cells are grown in a fixed volume of nutrient medium. No additional nutrients are added during cultivation, and the process continues until nutrients are depleted.
- ✓ ***Continuous culture:*** Fresh nutrient medium is continuously added while an equal amount of culture broth is removed, maintaining cells in an active growth phase.
- ✓ ***Fed-batch culture:*** Nutrients are added gradually during cultivation without removing the culture broth. It combines the advantages of batch and continuous culture and is widely used in industrial fermentation.

14.Write role of beta lactam antibiotics.

Beta-lactam antibiotics act by inhibiting the synthesis of bacterial cell walls. They bind to penicillin-binding proteins (PBPs) and block the formation of peptidoglycan, which is an essential component of the bacterial cell wall. As a result, the cell wall becomes weak and the bacteria undergo cell lysis and death. They are widely used for treating infections caused by Gram-positive and some Gram-negative bacteria. Examples include penicillin, ampicillin, and cephalosporins.

15.Write some uses of lactose and its limitations.

Uses of lactose:

- ✓ ***Food industry:*** Lactose is used as a sweetener and nutrient in dairy products, infant formulas, and bakery products.
- ✓ ***Pharmaceutical industry:*** It is used as a filler or carrier in tablets and capsules.
- ✓ ***Microbial fermentation:*** Lactose is used as a carbon source for the production of antibiotics, enzymes, and other microbial products.
- ✓ ***Dairy industry:*** It is used in the preparation of fermented dairy products such as yogurt.

Limitations of lactose:

- ✓ Lactose has low solubility compared with other sugars, which can limit its use in some processes.
- ✓ Many microorganisms cannot utilize lactose directly due to the absence of β-galactosidase enzyme.
- ✓ Some people are lactose intolerant because they cannot digest lactose properly.
- ✓ Its availability and cost may be higher compared with other carbon sources like glucose.

16.Why transposons are called jumping genes?

Transposons are called **"**jumping genes**"** because they can move or transfer from one location of the DNA to another within the genome. This movement is known as transposition. They can insert themselves into new DNA sites and may cause changes in gene expression, mutations, or alterations in genetic characteristics. This ability to change their position makes them different from ordinary genes.

17.Why biomass balance is important in medium of fermenter?

Biomass balance is important in the fermenter medium because it helps to maintain proper nutrient availability and optimize microbial growth**.** It provides information about the amount of biomass produced from the available nutrients and helps in calculating the consumption of carbon, nitrogen, and other essential components. A proper biomass balance helps to improve fermentation efficiency, control growth conditions, maximize product formation, and reduce waste of raw materials during the fermentation process.

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18.What is the meaning of attenuation in term of metabolic regulation?

Attenuation in metabolic regulation is a mechanism by which bacteria control gene expression by terminating transcription prematurely before the structural genes are fully transcribed. It occurs mainly in amino acid biosynthesis pathways, where the availability of the end product regulates enzyme production. When the required amino acid is abundant, transcription is stopped; when it is deficient, genes remain active to produce more amino acid. This helps the cell conserve energy and resources.

19.What are the byproducts in fermentation?

Byproducts in fermentation are the secondary products formed along with the desired fermentation product due to microbial metabolism. Common fermentation byproducts include:

- ✓ ***Alcohols:*** Ethanol, butanol, and glycerol.
- ✓ ***Organic acids:*** Lactic acid, acetic acid, citric acid, and succinic acid.
- ✓ ***Gases:*** Carbon dioxide and hydrogen.
- ✓ ***Solvents:*** Acetone and butanol.
- ✓ ***Other compounds:*** Vitamins, pigments, and flavoring compounds.

The type of byproduct produced depends on the microorganism, fermentation conditions, and the nutrients present in the medium.

20.Define two types of culture collection.

Two types of culture collection are:

- ✓ ***Public culture collection:*** These are organizations that maintain and distribute authenticated microbial strains for research, education, and industrial use. Examples include ATCC and DSMZ.
- ✓ ***Private culture collection:*** These are maintained by industries, research laboratories, or institutions to preserve specific microbial strains used for their own production processes and research purposes.

21.Write down the names of three industrial wastes.

Three industrial wastes are:

- ✓ ***Distillery waste (spent wash):*** Waste generated during alcohol production from fermentation processes.
- ✓ ***Dairy waste:*** Wastewater containing organic matter, proteins, fats, and lactose from dairy industries.
- ✓ ***Textile waste:*** Waste produced from textile processing industries containing dyes, chemicals, and other pollutants.

22.Write down the sources of organic and inorganic nitrogen.

Sources of organic nitrogen:

1. ***Peptone:*** A protein hydrolysate used as a rich nitrogen source in microbial growth media.
2. ***Yeast extract:*** Contains amino acids, peptides, and other nitrogenous compounds.
3. ***Soybean meal:*** A plant-based nitrogen source used in industrial fermentation.
4. ***Corn steep liquor:*** A byproduct of corn processing containing organic nitrogen compounds.

Sources of inorganic nitrogen:

1. ***Ammonium salts:*** Such as ammonium sulfate and ammonium chloride.
2. ***Nitrates:*** Such as sodium nitrate and potassium nitrate.
3. ***Ammonia:*** Used as a simple nitrogen source for microbial growth.

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23.What are trace elements? Give few examples.

Trace elements are essential mineral elements required by microorganisms in very small amounts for proper growth, enzyme activity, and metabolic functions. Although needed in low concentrations, they play important roles as enzyme cofactors and components of cellular structures. Examples of trace elements are: Iron (Fe), Zinc (Zn), Manganese (Mn), Copper (Cu), Cobalt (Co), Molybdenum (Mo) and Nickel (Ni).

24.Write names of five microorganisms in GRAS.

Five microorganisms included in GRAS (Generally Recognized As Safe) are:

1. ***Lactobacillus acidophilus*** – used in probiotics and fermented dairy products.
2. ***Bacillus subtilis*** – used in enzyme production and industrial fermentation.
3. ***Saccharomyces cerevisiae*** – yeast used in bread making and alcoholic fermentation.
4. ***Aspergillus oryzae*** – fungus used in the production of enzymes and fermented foods.
5. ***Lactococcus lactis*** – used in the dairy industry for cheese and yogurt production.

25.Write composition of malt extract.

Composition of malt extract:

Malt extract is commonly used as a growth medium for yeasts and fungi in fermentation studies. Malt extract is a nutrient-rich medium obtained from germinated barley and contains:

1. ***Carbohydrates (mainly maltose):*** Provides carbon and energy source for microbial growth.
2. ***Proteins and amino acids:*** Serve as a source of nitrogen for microorganisms.
3. ***Vitamins:*** Support microbial metabolism and growth.
4. ***Minerals and trace elements:*** Include phosphorus, potassium, magnesium, and iron.
5. ***Enzymes:*** Help in the breakdown of complex nutrients.

26.What are photoautotrophs? Explain its types.

Photoautotrophs:

Photoautotrophs are organisms that prepare their own food by using light energy and fixing carbon dioxide (CO_2) into organic compounds through photosynthesis. They use light as an energy source and inorganic substances as carbon sources.

Types of photoautotrophs:

1. ***Oxygenic photoautotrophs:*** These organisms perform photosynthesis by using water as an electron donor and release oxygen as a byproduct. ***Examples*:** Cyanobacteria, algae, and plants.

2. ***Anoxygenic photoautotrophs:*** These organisms perform photosynthesis without producing oxygen. They use substances like hydrogen sulfide (H_2S), hydrogen, or organic compounds as electron donors instead of water. ***Examples*:** Purple sulfur bacteria and green sulfur bacteria.

27.What is recombinant DNA?

Recombinant DNA is a type of DNA molecule formed by joining genetic material from two or more different sources using genetic engineering techniques. It is produced by inserting a desired gene into a vector, such as a plasmid, and introducing it into a host organism for expression. Recombinant DNA technology is used for the production of products like insulin, vaccines, hormones, and industrial enzymes.

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28.Differentiate between *E. coli* and *B. subtilis*.

Difference between *E. coli* and *B. subtilis*

E. coli	*B. subtilis*
It is a Gram-negative bacterium.	It is a Gram-positive bacterium.
It is a rod-shaped, non-spore-forming bacterium.	It is a rod-shaped, spore-forming bacterium.
It has an outer membrane containing lipopolysaccharides (LPS).	It lacks an outer membrane and contains a thick peptidoglycan layer.
It is commonly found in the intestinal tract of humans and animals.	It is commonly found in soil and plant environments.
It is widely used for recombinant protein production and genetic studies.	It is used for enzyme production and industrial fermentation.

29.Define inhibitor with examples.

An inhibitor is a substance that decreases or prevents the activity of an enzyme or slows down a biochemical reaction by interfering with microbial metabolism. In fermentation, inhibitors can reduce microbial growth and product formation. Examples are:

1. Heavy metals such as mercury and lead, which inhibit enzyme activity.
2. Organic acids such as acetic acid and lactic acid, which can inhibit microbial growth.
3. Antibiotics that prevent the growth of sensitive microorganisms.

LONG QUESTIONS

1. Explain the steps of Calvin cycle.

Calvin Cycle

The Calvin cycle is a series of light-independent reactions of photosynthesis that occur in the stroma of chloroplasts (in plants) and in the cytoplasm of photosynthetic bacteria. It uses ATP and NADPH produced during the light reaction to convert carbon dioxide (CO_2) into carbohydrates. The Calvin cycle consists of three main steps:

1. Carbon Fixation

- In this step, carbon dioxide (CO_2) is attached to a five-carbon compound called ribulose-1,5-bisphosphate (RuBP).
- The reaction is catalyzed by the enzyme RuBisCO (ribulose bisphosphate carboxylase/oxygenase).
- The unstable six-carbon compound formed immediately breaks into two molecules of 3-phosphoglycerate (3-PGA).

2. Reduction Phase

- In this stage, 3-PGA is converted into glyceraldehyde-3-phosphate (G3P).
- This conversion requires energy from ATP and reducing power from NADPH produced during the light-dependent reactions.
- Some G3P molecules leave the cycle and are used to form carbohydrates such as glucose.

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3. Regeneration of RuBP

- The remaining G3P molecules are used to regenerate RuBP, the CO_2 acceptor molecule.
- This step requires ATP energy to reform RuBP so that the cycle can continue.
- The regenerated RuBP allows continuous fixation of more carbon dioxide.

Importance of Calvin Cycle:

- It converts inorganic carbon dioxide into organic compounds.
- It produces carbohydrates that serve as an energy source for organisms.
- It plays an essential role in carbon fixation and maintaining the carbon cycle in nature.

2. What factors influence oxygen provision in fermenter?

Factors Influencing Oxygen Provision in Fermenter

Oxygen provision is an important factor in aerobic fermentation because microorganisms require oxygen for growth, metabolism, and product formation. The transfer of oxygen from air bubbles into the fermentation broth depends on several factors:

1. Agitation Rate

Agitation increases the mixing of the fermentation broth and breaks air bubbles into smaller sizes. Smaller bubbles provide a larger surface area for oxygen transfer. Higher agitation generally increases oxygen availability to microorganisms.

2. Aeration Rate

Aeration is the supply of sterile air into the fermenter. Increasing the airflow rate increases the amount of oxygen available for microbial respiration. Proper aeration prevents oxygen limitation during fermentation.

3. Oxygen Solubility in Medium

The amount of oxygen dissolved in the fermentation broth affects oxygen availability. Factors such as temperature, pressure, and composition of the medium influence oxygen solubility.

4. Viscosity of Fermentation Broth

Highly viscous media reduce the movement of oxygen bubbles and decrease oxygen transfer efficiency. Substances such as polysaccharides and high cell concentrations can increase viscosity.

5. Temperature

Temperature affects both microbial oxygen demand and oxygen solubility. Higher temperatures generally decrease oxygen solubility in the broth.

6. Pressure

Increasing the pressure inside the fermenter increases oxygen solubility and improves oxygen transfer.

7. Fermenter Design

The shape and size of the fermenter, type of impeller, and placement of spargers influence oxygen mixing and transfer efficiency.

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8. Microbial Cell Concentration

High cell density increases oxygen consumption. If oxygen supply is lower than oxygen demand, microbial growth and product formation may decrease.

3. Why foam is produced during fermentation? What problem it causes?

Foam Formation During Fermentation and Its Problems

Foam is commonly produced during fermentation due to the interaction of air, agitation, and surface-active substances present in the fermentation medium. During aeration, air bubbles are introduced into the broth, and substances such as proteins, peptides, amino acids, and carbohydrates stabilize these bubbles, resulting in foam formation.

Causes of Foam Production:

- ✓ ***Aeration:*** Continuous supply of air creates bubbles in the fermentation broth.
- ✓ ***Agitation:*** Mixing increases the formation and stability of air bubbles.
- ✓ ***Presence of proteins and organic compounds:*** These compounds reduce surface tension and stabilize foam.
- ✓ ***High microbial growth:*** Increased cell mass and metabolic products can enhance foam formation.

Problems Caused by Excessive Foam:

- ✓ ***Contamination risk:*** Foam may reach the top of the fermenter and cause contamination through filters or openings.
- ✓ ***Loss of culture medium:*** Excess foam can cause overflow, leading to loss of valuable fermentation broth.
- ✓ ***Reduced oxygen transfer:*** Excess foam can interfere with proper gas exchange and oxygen availability.
- ✓ ***Interference with sensors:*** Foam can affect pH, temperature, and oxygen sensors, causing inaccurate readings.
- ✓ ***Reduced fermenter efficiency:*** It decreases the effective working volume and may reduce product yield.
- ✓ ***Blockage of air outlets:*** Excessive foam can block exhaust filters and disturb pressure control.

4. Why foam is produced during fermentation? Write approaches how to control it.

Foam Formation During Fermentation and Approaches to Control It

Foam is produced during fermentation due to the presence of air bubbles, agitation, and surface-active substances in the fermentation broth. Aeration introduces air into the medium, and compounds such as proteins, peptides, carbohydrates, and microbial metabolites stabilize the bubbles, resulting in foam formation. High cell concentration and vigorous mixing can also increase foam production.

Causes of Foam Formation:

1. ***Aeration:*** Continuous supply of air creates many bubbles in the broth.
2. ***Agitation:*** Mechanical stirring increases bubble formation and dispersion.
3. ***Surface-active compounds:*** Proteins and other organic molecules reduce surface tension and stabilize foam.
4. ***Microbial growth:*** Growing cells release substances that promote foam stability.

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Approaches to Control Foam:

- ✓ ***Use of Antifoaming Agents:*** Chemical antifoams such as silicone oils, mineral oils, and vegetable oils are added to break foam bubbles. They reduce surface tension and prevent excessive foam formation.
- ✓ ***Mechanical Foam Control:*** Mechanical foam breakers or foam disruptors can be installed in fermenters to physically destroy foam.
- ✓ ***Control of Aeration Rate:*** Reducing excessive airflow helps decrease bubble formation and limits foam production.
- ✓ ***Control of Agitation Speed:*** Maintaining an appropriate agitation rate prevents unnecessary foam generation while ensuring proper mixing.
- ✓ ***Modification of Medium Composition:*** Reducing the concentration of foam-producing nutrients, especially proteins and peptides, can minimize foam formation.
- ✓ ***Proper Fermenter Design:*** Use of suitable vessel design, impellers, and foam sensors helps maintain foam at controlled levels.

5. What characteristics should be present in antifoaming agent?

Characteristics of an Ideal Antifoaming Agent

An antifoaming agent is a substance added to fermentation media to control excessive foam formation. An ideal antifoaming agent should have the following characteristics:

- ✓ ***High foam-breaking ability:*** It should quickly break existing foam bubbles and prevent the formation of new foam.
- ✓ ***Low toxicity:*** It should not harm microbial cells or interfere with their growth and metabolism.
- ✓ ***Chemically stable:*** It should remain stable under fermentation conditions such as changes in pH, temperature, and pressure.
- ✓ ***Inert nature:*** It should not react with nutrients, fermentation products, or other components of the medium.
- ✓ ***Effective at low concentration:*** A small amount of antifoam should be sufficient to control foam, making it economical.
- ✓ ***Easy to disperse in the medium:*** It should mix properly with the fermentation broth and spread throughout the fermenter.
- ✓ ***Should not affect oxygen transfer:*** It should control foam without greatly reducing oxygen availability to microorganisms.
- ✓ ***Easy removal during downstream processing:*** It should not interfere with product recovery or purification processes.
- ✓ ***Sterilization stability:*** It should tolerate sterilization conditions without losing its effectiveness.

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6. Which microorganisms are used in fermentation of products process?

Microorganisms Used in Fermentation Product Processes

A variety of microorganisms are used in industrial fermentation for the production of alcohols, organic acids, antibiotics, enzymes, vitamins, amino acids, and other valuable products. The choice of microorganism depends on the type of product required.

1. Bacteria

- *Corynebacterium glutamicum* – used for the production of glutamic acid and amino acids.
- *Lactobacillus* species – used for the production of lactic acid and fermented dairy products.
- *Bacillus subtilis* – used for the production of enzymes and antibiotics.
- *Acetobacter aceti* – used for the production of acetic acid (vinegar).

2. Yeasts

- *Saccharomyces cerevisiae* – used in the production of ethanol, bread, and alcoholic beverages.
- *Candida utilis* – used for single-cell protein production and nutritional products.

3. Fungi

- *Aspergillus niger* – used for the production of citric acid and enzymes.
- *Aspergillus oryzae* – used in food fermentation and enzyme production.
- *Penicillium chrysogenum* – used for the production of penicillin.

4. Actinomycetes

- *Streptomyces griseus* – used for the production of streptomycin antibiotic.
- *Streptomyces* species – widely used for various antibiotic productions.

7. Enlist attributes for an ideal industrial microorganism.

Attributes of an Ideal Industrial Microorganism

An ideal industrial microorganism should possess characteristics that make it suitable for large-scale fermentation and efficient product formation. Important attributes include:

- ✓ ***High productivity:*** The microorganism should produce a large amount of the desired product in a short period of time.
- ✓ ***Rapid growth rate:*** It should grow quickly and produce sufficient biomass or metabolites for industrial use.
- ✓ ***Genetic stability:*** The strain should maintain its desirable characteristics and product-forming ability over many generations.
- ✓ ***Non-pathogenic nature:*** It should be safe to handle and should not produce harmful toxins or cause disease.
- ✓ ***Ability to utilize inexpensive raw materials:*** It should grow on low-cost substrates such as agricultural wastes or industrial byproducts.
- ✓ ***Tolerance to environmental conditions:*** It should withstand changes in pH, temperature, osmotic pressure, and high product concentration.
- ✓ ***Easy recovery of product:*** The microorganism should produce products that can be easily extracted and purified.

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- ✓ ***Production of few unwanted byproducts:*** It should produce minimal unwanted metabolites to simplify downstream processing.
- ✓ ***Ability to grow in large-scale fermenters:*** It should perform efficiently under industrial fermentation conditions.
- ✓ ***High yield of desired metabolite:*** The microorganism should convert nutrients efficiently into the required product.

8. What factors affect performance of fermenter?

Factors Affecting Performance of Fermenter

The performance of a fermenter depends on various physical, chemical, and biological factors that influence microbial growth and product formation.

1. Temperature

Temperature affects enzyme activity, microbial growth, and metabolic reactions. Each microorganism has an optimum temperature range for maximum product formation. Extreme temperatures can reduce growth or kill the cells.

2. pH Control

pH influences microbial metabolism, enzyme activity, and nutrient availability. Maintaining the correct pH is essential for proper growth and efficient product production.

3. Oxygen Supply and Aeration

Aerobic microorganisms require sufficient oxygen for respiration and growth. Proper aeration ensures adequate oxygen transfer from air bubbles to the fermentation broth.

4. Agitation

Agitation provides uniform mixing of nutrients, cells, and oxygen throughout the fermenter. It improves oxygen transfer and prevents cell settling.

5. Nutrient Composition of Medium

The availability of carbon sources, nitrogen sources, minerals, and vitamins affects microbial growth and product yield. An unsuitable medium can decrease fermentation efficiency.

6. Inoculum Size and Quality

The amount and health of the starter culture influence fermentation performance. A suitable inoculum size leads to faster growth and better product formation.

7. Foam Formation

Excessive foam can interfere with oxygen transfer, cause contamination, and reduce working volume. Foam control is necessary for efficient operation.

8. Sterility Conditions

Contamination by unwanted microorganisms can compete for nutrients and reduce product quality. Maintaining sterile conditions is essential.

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9. Pressure and Dissolved Oxygen Level

Pressure affects oxygen solubility, while dissolved oxygen concentration influences microbial activity. Proper control improves fermentation efficiency.

10. Fermenter Design

The size, shape, impeller type, and aeration system of the fermenter affect mixing, heat transfer, and oxygen supply.

9. Explain transduction and its process.

Transduction and Its Process

Transduction is a mechanism of horizontal gene transfer in bacteria in which genetic material is transferred from one bacterium to another by a bacteriophage (bacterial virus)**.** During this process, the bacteriophage carries DNA from a donor bacterial cell and introduces it into a recipient bacterial cell.

Types of Transduction:

- ✓ ***Generalized Transduction:*** In this type, a bacteriophage can transfer any part of bacterial DNA from one cell to another. It occurs when a phage accidentally packages bacterial DNA instead of its own genetic material.

- ✓ ***Specialized Transduction:*** In this type, only specific bacterial genes located near the phage integration site are transferred. It occurs when a temperate phage incorrectly removes bacterial genes along with its own DNA.

Process of Transduction:

- ✓ ***Attachment of Phage:*** A bacteriophage attaches to the surface of a donor bacterial cell.
- ✓ ***Injection of Phage DNA:*** The phage injects its genetic material into the bacterial cell.
- ✓ ***Replication and DNA Packaging:*** The phage replicates inside the host cell.
- ✓ During assembly, some phage particles may accidentally contain bacterial DNA.
- ✓ ***Release of Phage Particles:*** The infected bacterial cell breaks open, releasing new phage particles carrying bacterial DNA.
- ✓ ***Infection of Recipient Cell:*** The modified phage attaches to another bacterial cell and transfers the carried bacterial DNA.
- ✓ ***Integration of DNA:*** The transferred DNA recombines with the recipient cell's chromosome, producing new genetic characteristics.

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10.Briefly explain shotgun and objective approach.

Shotgun and Objective Approach

1. Shotgun Approach:

The shotgun approach is a method used in genetic engineering and genome analysis in which the entire DNA of an organism is randomly broken into many small fragments and then analyzed or sequenced.

Steps involved:

1. The complete genomic DNA is isolated from the organism.
2. DNA is randomly cut into small fragments.
3. The fragments are cloned into suitable vectors.
4. The DNA fragments are sequenced and analyzed.
5. The complete genome sequence is reconstructed using computer-based methods.

Advantages:

- Faster method for analyzing large genomes.
- Does not require prior knowledge of the DNA sequence.
- Useful for discovering new genes and metabolic pathways.

Limitations:

- Requires advanced computational tools for sequence assembly.
- Difficult to analyze repetitive DNA regions.

2. Objective (Targeted) Approach:

The objective approach is a specific method in which a particular gene or DNA sequence is selected and studied based on a known function or desired characteristic.

Steps involved:

1. A specific gene of interest is identified.
2. The gene is isolated using molecular techniques.
3. Its structure and function are analyzed.
4. The gene may be modified or transferred into a suitable host.

Advantages:

- More precise and focused than the shotgun approach.
- Requires less data analysis.
- Useful for studying specific genes and producing desired products.

Limitations:

- Requires prior knowledge of the gene or its function.
- Cannot easily identify unknown genes.

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11.What is anoxygenic photosynthetic bacteria?

Anoxygenic Photosynthetic Bacteria

Anoxygenic photosynthetic bacteria are microorganisms that perform photosynthesis without producing oxygen as a byproduct. Unlike plants and cyanobacteria, they do not use water as an electron donor. Instead, they use substances such as hydrogen sulfide (H_2S), hydrogen (H_2), or organic compounds as electron donors during photosynthesis. These bacteria contain special photosynthetic pigments such as bacteriochlorophyll and use light energy to convert carbon dioxide into organic compounds.

Types and Examples:

- ✓ ***Purple sulfur bacteria:*** They use hydrogen sulfide (H_2S) as an electron donor. They store sulfur granules inside or outside the cells. **Example**: *Chromatium*.
- ✓ ***Purple non-sulfur bacteria:*** They use organic compounds or hydrogen as electron donors.
- ✓ Commonly found in aquatic environments. **Example**: *Rhodospirillum*.
- ✓ ***Green sulfur bacteria:*** They use hydrogen sulfide as an electron donor and deposit sulfur outside the cells. **Example**: *Chlorobium*.

12.Explain three kingdoms of Archaea.

Three Kingdoms of Archaea

Archaea are a group of unicellular prokaryotic microorganisms that differ from bacteria in their cell wall structure, membrane composition, and genetic characteristics. They are often found in extreme environments such as high temperature, high salt, and anaerobic conditions. Archaea are divided into three main kingdoms:

1. Euryarchaeota

- ✓ This is the largest and most diverse group of Archaea.
- ✓ Members include methanogens and extreme halophiles.
- ✓ Methanogens produce methane gas as a metabolic end product and are found in anaerobic environments such as swamps and animal digestive systems.
- ✓ Halophiles survive in highly salty environments.
- ✓ **Examples:** *Methanobacterium*, *Halobacterium*.

2. Crenarchaeota

- ✓ These Archaea are mainly found in extreme environments, especially high-temperature and acidic habitats.
- ✓ Many are thermophiles and acidophiles that can survive harsh conditions.
- ✓ They are commonly found in hot springs, volcanic areas, and deep-sea hydrothermal vents.
- ✓ **Examples:** *Sulfolobus*, *Thermoproteus*.

3. Korarchaeota

- ✓ This is a less-studied group of Archaea found mainly in hot environments.
- ✓ They are considered ancient forms of Archaea with unique genetic features.
- ✓ They have been identified in hot springs and geothermal habitats.
- ✓ **Example:** *Korarchaeum cryptofilum*.

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13.Write limitations of prokaryotes in expression of gene.

Limitations of Prokaryotes in Gene Expression

Prokaryotic organisms such as bacteria are widely used for gene expression, but they have several limitations when producing complex proteins, especially eukaryotic proteins.

1. Lack of Post-Translational Modifications

Prokaryotes cannot perform many post-translational modifications such as glycosylation, phosphorylation, and protein folding required for proper function of some eukaryotic proteins. As a result, some human proteins produced in bacteria may not be biologically active.

2. Formation of Inclusion Bodies

Foreign proteins expressed at high levels in prokaryotes may accumulate as insoluble aggregates called inclusion bodies. These proteins require additional steps for purification and refolding.

3. Absence of Introns Removal Mechanism

Prokaryotes lack the machinery required for removing introns from eukaryotic genes. Therefore, genes containing introns must be modified into intron-free forms before expression.

4. Improper Protein Folding

Some complex proteins require specific cellular conditions and molecular chaperones for correct folding. Prokaryotic cells may not provide suitable folding mechanisms.

5. Limited Secretion Ability

Prokaryotes have limited ability to secrete recombinant proteins outside the cell. This makes product recovery more difficult.

6. Differences in Regulatory Systems

Prokaryotic gene regulation systems differ from eukaryotic systems. Foreign genes may not be expressed efficiently due to differences in promoters and regulatory elements.

7. Proteolytic Degradation

Some recombinant proteins may be degraded by bacterial proteases, reducing product yield.

8. Endotoxin Production

Gram-negative bacteria such as *E. coli* produce endotoxins (lipopolysaccharides) that must be removed from pharmaceutical products.

14.Give two theories of floc formation.

Theories of Floc Formation

Floc formation is the process in which microbial cells aggregate together to form clusters called flocs. It is important in fermentation and wastewater treatment because it improves cell separation and settling. Two main theories explain floc formation:

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1. Bridging Theory

According to this theory, floc formation occurs when polymeric substances produced by microorganisms act as bridges between individual cells. These polymers, such as extracellular polysaccharides, attach to the surfaces of different cells and bind them together. This results in the formation of stable microbial aggregates or flocs.

2. Polymer Entrapment Theory

This theory suggests that microbial cells become trapped within a network of extracellular polymers. The sticky polymer matrix holds the cells together and forms a floc structure. The strength and size of flocs depend on the amount and nature of extracellular polymeric substances.

15.How heat influences the microbial growth?

Influence of Heat on Microbial Growth

Temperature is one of the most important environmental factors affecting microbial growth. Heat influences microbial growth by affecting enzyme activity, cellular structure, metabolism, and survival of microorganisms. Each microorganism has an optimum temperature range at which it grows best.

1. Effect of Low Temperature

- ✓ Low temperatures reduce enzyme activity and slow down microbial metabolism.
- ✓ Nutrient transport and biochemical reactions become slower, resulting in reduced growth rate.
- ✓ Some microorganisms can survive at low temperatures but grow slowly.

2. Effect of Optimum Temperature

- ✓ At the optimum temperature, enzymes function efficiently and metabolic activities occur at maximum rates.
- ✓ Microorganisms show the highest growth rate and maximum product formation.
- ✓ Industrial fermentation processes are usually maintained at the optimum temperature for the selected microorganism.

3. Effect of High Temperature

- ✓ High temperatures denature proteins and enzymes, causing loss of cellular functions.
- ✓ Excessive heat damages cell membranes and nucleic acids, leading to cell death.
- ✓ Heat can also destroy important cellular structures.

Classification of Microorganisms Based on Temperature:

- ✓ ***Psychrophiles:*** Grow best at low temperatures (approximately 0–20°C). **Example**: Cold-environment bacteria.
- ✓ ***Mesophiles:*** Grow best at moderate temperatures (approximately 20–45°C). Most industrial microorganisms belong to this group. **Example**: *Escherichia coli* and *Saccharomyces cerevisiae*.
- ✓ ***Thermophiles:*** Grow at high temperatures (approximately 45–80°C). **Example**: *Thermus aquaticus*.
- ✓ ***Hyperthermophiles:*** Grow at extremely high temperatures above 80°C. **Example**: Some archaeal species.

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16.Explain structure of fungal hyphae.

Structure of Fungal Hyphae

Hyphae are long, thread-like tubular structures that form the main body of fungi. A network of hyphae is called a mycelium, which helps fungi grow, absorb nutrients, and reproduce.

Structure of Hyphae:

- ✓ ***Cell Wall:*** The outer covering of hyphae is made mainly of chitin and glucans. It provides strength, protection, and maintains the shape of the fungal cell.
- ✓ ***Plasma Membrane:*** Located beneath the cell wall. Controls the movement of nutrients and other substances into and out of the fungal cell.
- ✓ ***Cytoplasm:*** The internal material of hyphae contains organelles such as mitochondria, ribosomes, and vacuoles. It supports metabolic activities and growth.
- ✓ ***Nuclei:*** Hyphae may contain one or more nuclei depending on the fungal species. Multinucleate hyphae are common in many fungi.
- ✓ ***Septae:*** Some hyphae are divided by cross walls called septa. Septa contain pores that allow movement of cytoplasm and organelles between compartments.
- ✓ ***Apical Region:*** The tip of the hypha is called the apical region. It is the active site of growth where new cell material is added.

17.Explain process of amino acid catabolism.

Process of Amino Acid Catabolism

Amino acid catabolism is the process by which amino acids are broken down to produce energy, metabolic intermediates, and nitrogen-containing compounds. It occurs when amino acids are used as an energy source or when they are present in excess. The process involves three main steps:

1. Deamination

- ✓ In this step, the amino group ($-NH_2$) is removed from the amino acid.
- ✓ This reaction is carried out by enzymes called deaminases.
- ✓ The amino group is converted into ammonia (NH_3) or ammonium ions (NH_4^+).
- ✓ The remaining carbon skeleton is converted into keto acids, which can enter metabolic pathways.
- ✓ ***Example*:** Glutamate → α-ketoglutarate + ammonia

2. Transamination

- ✓ Transamination is the transfer of an amino group from one amino acid to another molecule.
- ✓ The reaction is catalyzed by enzymes called aminotransferases (transaminases)**.**
- ✓ It helps in the formation of new amino acids and prepares amino acids for further breakdown.

3. Breakdown of Carbon Skeleton

- ✓ After removal of the amino group, the remaining carbon skeleton is converted into compounds that enter central metabolic pathways.
- ✓ These compounds may enter the Krebs cycle (TCA cycle) to produce energy in the form of ATP.
- ✓ Some carbon skeletons can also be used for the synthesis of glucose or other cellular components.

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Fate of Nitrogen:

- ✓ The released ammonia is toxic and must be removed or converted into less harmful compounds.
- ✓ Microorganisms may use ammonia for nitrogen assimilation or release it as a waste product.

18.Define beer. What is the objective of mashing process?

Beer and Objective of Mashing Process

Beer:

Beer is an alcoholic fermented beverage produced from malted grains, mainly barley, through the action of yeast. During fermentation, yeast (commonly *Saccharomyces cerevisiae*) converts fermentable sugars into ethanol and carbon dioxide, along with various flavor compounds. Beer contains water, alcohol, carbon dioxide, carbohydrates, proteins, and flavoring substances.

Objective of Mashing Process:

The mashing process is an important step in beer production in which crushed malted grains are mixed with warm water to activate enzymes present in the malt. The main objectives of mashing are:

- ✓ ***Conversion of starch into fermentable sugars:*** Enzymes such as amylase break down starch into sugars like maltose and glucose, which can be fermented by yeast.
- ✓ ***Extraction of soluble nutrients:*** It releases proteins, minerals, vitamins, and other nutrients required for yeast growth and fermentation.
- ✓ ***Preparation of wort:*** The liquid extract obtained after mashing, called wort, contains fermentable sugars needed for alcohol production.
- ✓ ***Development of flavor and composition:*** Mashing helps produce the desired taste, color, and quality characteristics of beer.

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